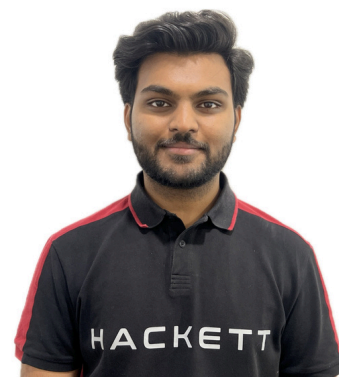


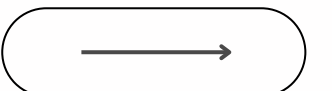
DATE
06/01/2025



FACILITY MANAGEMENT SYSTEM

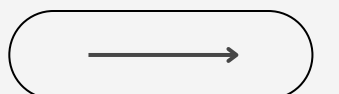


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Problem Statement

- **Objective:** Create a system for seamless room booking, guest management, and service coordination.
- Challenges:
 - Manual processes leading to inefficiencies.
 - Delays in communication between departments.
 - Need for secure handling of guest details.



KEY FEATURES & WORKFLOW

Room management for multiple categories:

- Deluxe, Luxury, and Luxury Suite (across 10 floors).

Automated notifications for:

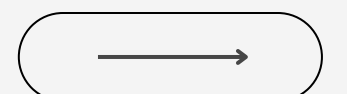
- Room service.
- WiFi credentials.
- Restaurant menu updates.

Real-time updates on:

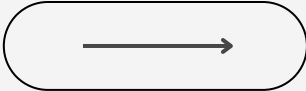
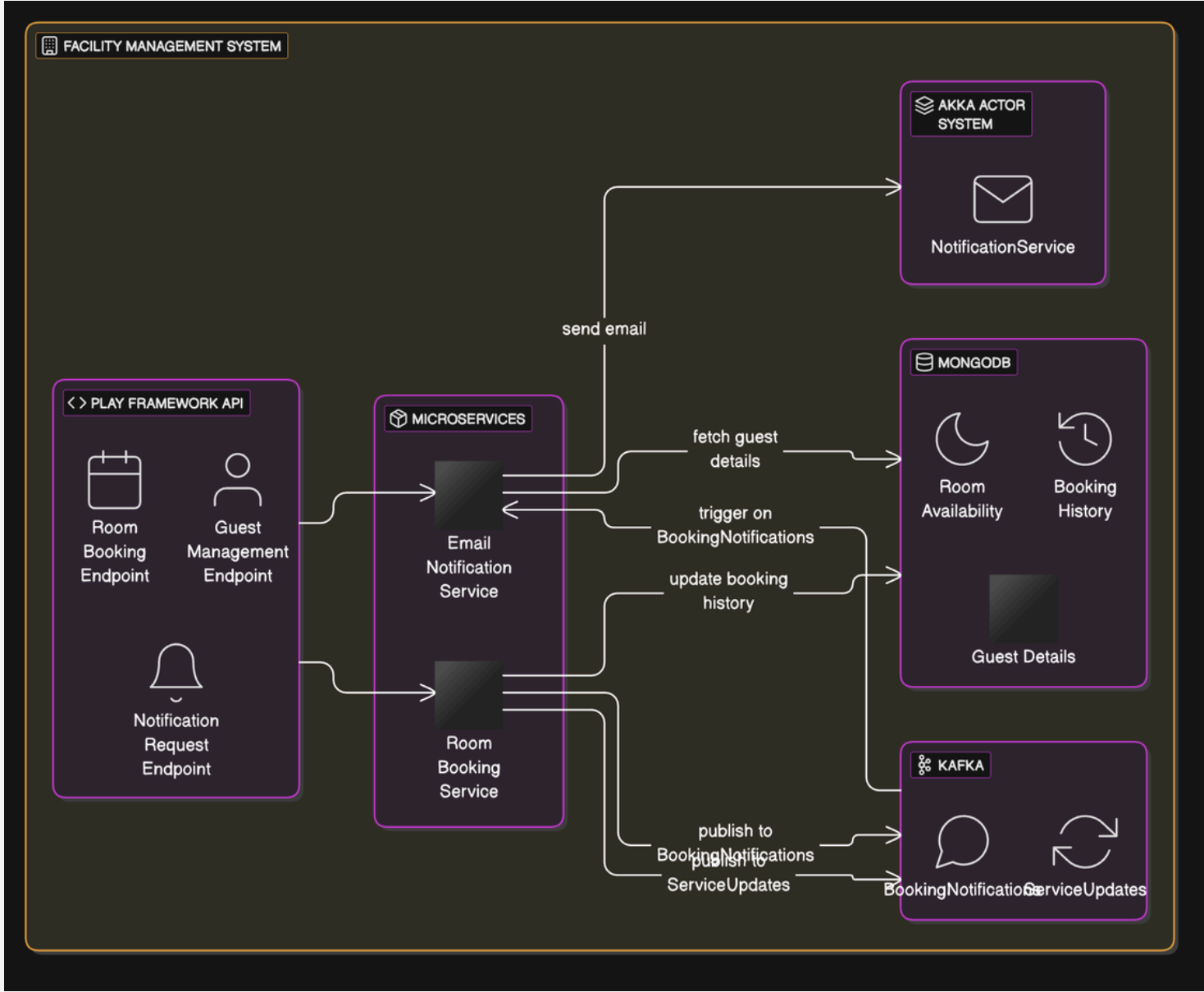
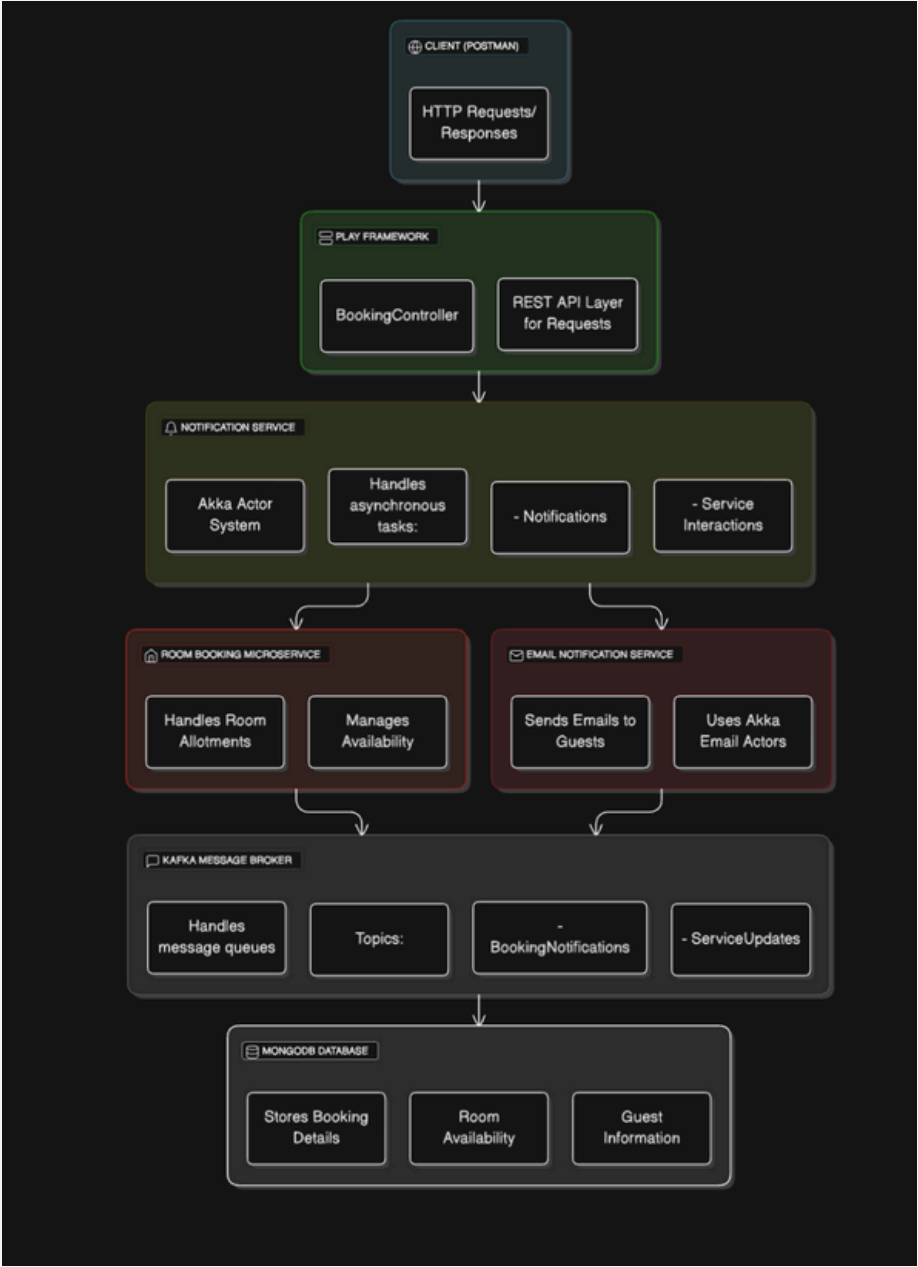
- Room availability & Booking history.
- Secure data handling:
- Guest details and ID proof uploads & Micro services for efficient communication.

Architecture: Modular microservices.

Communication: Asynchronous, powered by Kafka and Akka.



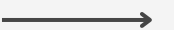
SYSTEM ARCHITECTURE



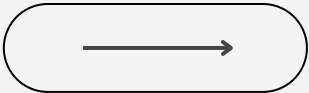
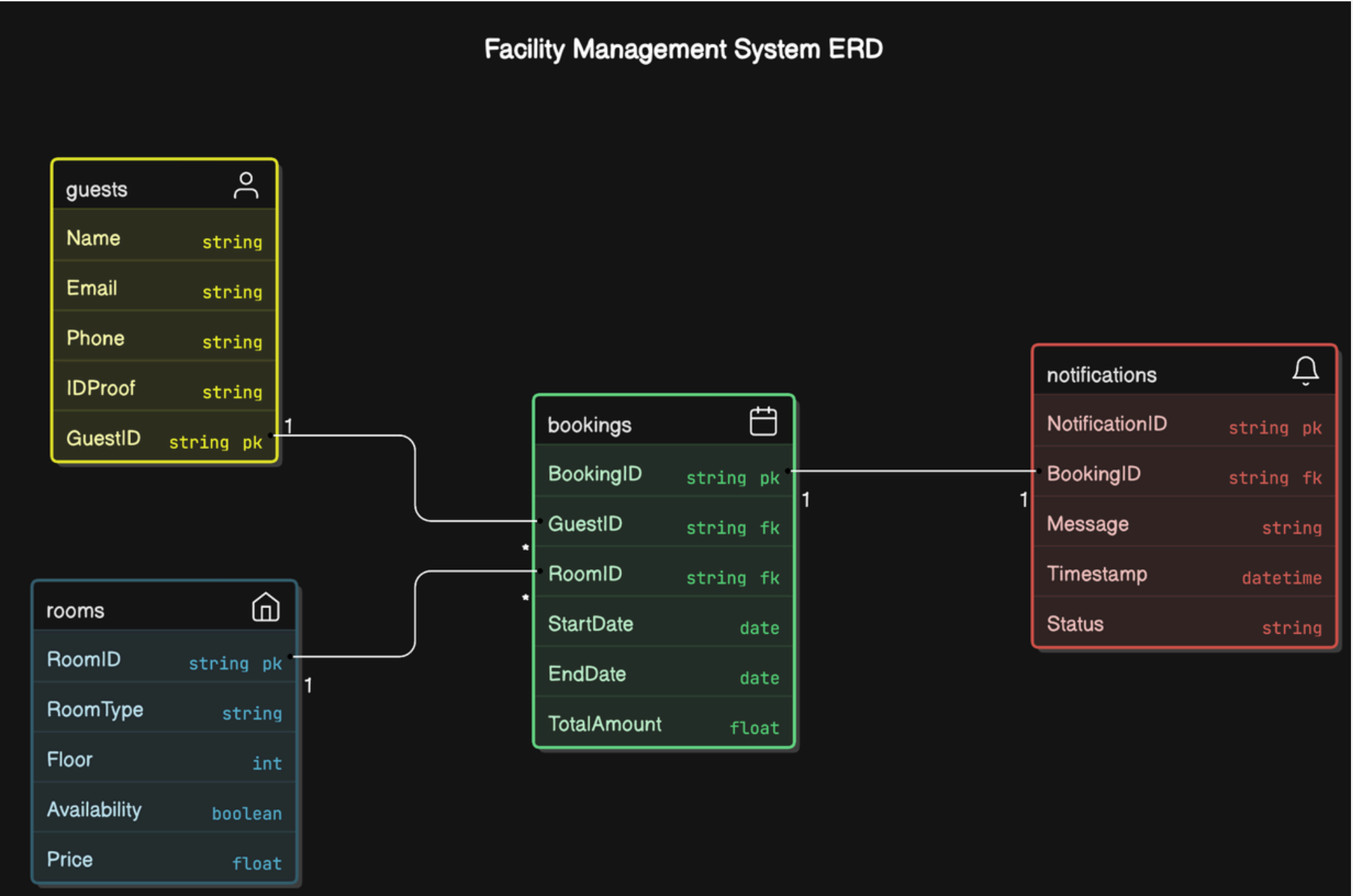
TECHNOLOGIES AND KEY COMPONENTS

- Technology Stack
- Backend Framework: Play Framework (Scala)
- Asynchronous Tasks: Akka Actor System
- Messaging: Kafka
- Database: MongoDB
- Build Tool: SBT
- Containerization: Docker
- Testing: ScalaTest
- Email Notifications: Akka Email Actors

- Play Framework API:
- Handles REST APIs for bookings and guest management.
- Routes requests to microservices.
- Akka Actor System:
- Manages async tasks.
- NotificationService: Automated email dispatch.
- Kafka Message Broker:
 - Topics: BookingNotifications, ServiceUpdates.
- MongoDB Database:
- Stores guest details, room availability, and booking history.
- Encrypts sensitive guest data.
- Microservices:
- Room Booking Service.
- Email Notification Service.



DATA SCHEMA



APIs

BOOKINGS

1. Get All Bookings (GET /bookings)

Response: List of all bookings

2. Get Booking By ID (GET /bookings/{id})

Response: Booking details

3. Create Booking (POST /bookings)

Request: { id, guestId, roomId, checkInDate, checkOutDate }

Response: Booking created confirmation

4. Delete Booking (DELETE /bookings/{id})

Response: Booking canceled confirmation

GUESTS

1. Get All Guests (GET /guests)

Response: List of all guests

2. Get Guest By ID (GET /guests/{id})

Response: Guest details

3. Add Guest (POST /guests)

Request: { id, name, contact, email }

Response: Guest added confirmation

4. Delete Guest (DELETE /guests/{id})

Response: Guest deleted confirmation

NOTIFICATIONS

1. Get All Notifications (GET /notifications)

Response: List of all notifications

2. Send Notification (POST /notifications)

Request: { id, recipient, message, timestamp, status }

Response: Notification sent confirmation

ROOMS

1. Get All Rooms (GET /rooms)

Response: List of all rooms

2. Get Room By ID (GET /rooms/{id})

Response: Room details

3. Add Room (POST /rooms)

Request: { id, type, status, price }

Response: Room added confirmation

4. Delete Room (DELETE /rooms/{id})

Response: Room deleted confirmation

API Responses

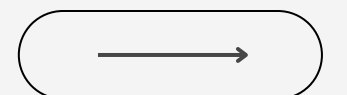
```
{
  "booking": {
    "getAllBookings": [
      {
        "id": "1",
        "guestId": "101",
        "roomId": "201",
        "checkInDate": "2025-01-06",
        "checkOutDate": "2025-01-07",
        "status": "Confirmed"
      }
    ],
    "getBookingById": {
      "id": "1",
      "guestId": "101",
      "roomId": "201",
      "checkInDate": "2025-01-06",
      "checkOutDate": "2025-01-07",
      "status": "Confirmed"
    },
    "createBooking": {
      "message": "Booking created successfully"
    },
    "deleteBooking": {
      "message": "Booking canceled"
    }
  },
  "guest": {
    "getAllGuests": [
      {
        "id": "101",
        "name": "John Doe",
        "contact": "1234567890",
        "email": "john.doe@example.com"
      }
    ],
    "getGuestById": {
      "id": "101",
      "name": "John Doe",
      "contact": "1234567890",
      "email": "john.doe@example.com"
    },
    "addGuest": {
      "message": "Guest added successfully"
    },
    "deleteGuest": {
      "message": "Guest deleted"
    }
  },
  "notification": {
    "getAllNotifications": [
      {
        "id": "301",
        "recipient": "John Doe",
        "message": "Your booking is confirmed.",
        "timestamp": "2025-01-05T10:00:00Z",
        "status": "Sent"
      }
    ],
    "sendNotification": {
      "message": "Notification sent"
    }
  },
  "room": {
    "getAllRooms": [
      {
        "id": "201",
        "type": "Deluxe",
        "status": "Available",
        "price": 150
      }
    ],
    "getRoomById": {
      "id": "201",
      "type": "Deluxe",
      "status": "Available",
      "price": 150
    },
    "addRoom": {
      "message": "Room added successfully"
    },
    "deleteRoom": {
      "message": "Room deleted"
    }
  }
}
```

MONGODB Collection JSON

```
{
  "bookings": [
    {
      "_id": "1",
      "guestId": "101",
      "roomId": "201",
      "checkInDate": "2025-01-06",
      "checkOutDate": "2025-01-07",
      "status": "Confirmed"
    },
    {
      "_id": "2",
      "guestId": "102",
      "roomId": "202",
      "checkInDate": "2025-01-08",
      "checkOutDate": "2025-01-10",
      "status": "Pending"
    }
  ],
  "guests": [
    {
      "_id": "101",
      "name": "John Doe",
      "contact": "1234567890",
      "email": "john.doe@example.com"
    },
    {
      "_id": "102",
      "name": "Jane Smith",
      "contact": "9876543210",
      "email": "jane.smith@example.com"
    }
  ],
  "notifications": [
    {
      "_id": "301",
      "recipient": "John Doe",
      "message": "Your booking is confirmed.",
      "timestamp": "2025-01-05T10:00:00Z",
      "status": "Sent"
    },
    {
      "_id": "302",
      "recipient": "Jane Smith",
      "message": "Your room is ready.",
      "timestamp": "2025-01-06T09:30:00Z",
      "status": "Pending"
    }
  ],
  "rooms": [
    {
      "_id": "201",
      "type": "Deluxe",
      "status": "Available",
      "price": 150
    },
    {
      "_id": "202",
      "type": "Suite",
      "status": "Occupied",
      "price": 250
    }
  ]
}
```

FUTURE ENHANCEMENTS

- Guest self-service mobile app.
- AI-driven service recommendations.
- Multi-property management capabilities.



Thank You